impl PrivacyProfile::from_pairs

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This proof resides in **"contrib"** because it has not completed the vetting process.

1 Hoare Triple

Precondition

Compiler-verified

- Argument `pairs` of type `Vec<(f64, f64)>`

Caller-verified

None

Pseudocode

```python
import bisect

def privacyprofile_from_pairs(pairs: list[tuple[float, float]]) -> PrivacyProfile:
    if not pairs:
        raise ValueError("privacy profile must be defined by at least one pair")

    # Validate pairs
    for eps, delta in pairs:
        if not (isinstance(eps, float) and eps >= 0 and eps != float('inf')):
            raise ValueError("epsilon must be finite and non-negative")
        if not (isinstance(delta, float) and 0 <= delta <= 1):
            raise ValueError("delta must be finite and in [0, 1]")

    # Sort increasing on epsilon, then decreasing on delta for tie-breaking
    pairs.sort(key=lambda x: (x[0], -x[1]))  #

    # For duplicate epsilons, keep the largest delta conservatively
    deduplicated = []  #
    for eps, delta in pairs:
        if not deduplicated or deduplicated[-1][0] != eps:
            deduplicated.append((eps, delta))
        else:
            # Keep the maximum delta for duplicate epsilons
            deduplicated[-1] = (eps, max(deduplicated[-1][1], delta))

    pairs = deduplicated  #
    epsilon_grid, delta_grid = zip(*pairs)
    epsilon_grid = list(epsilon_grid)
    delta_grid = list(delta_grid)
```

```
30
31      # Check that delta is monotonically nonincreasing
32      for window in zip(delta_grid, delta_grid[1:]):  #
33          if window[0] < window[1]:
34              raise ValueError(
35                  "delta values must be monotonically nonincreasing as epsilon increases"
36              )
37
38      def profile(epsilon: float) -> float:
39          if not (isinstance(epsilon, float) and epsilon >= 0 and epsilon == epsilon):
40              raise ValueError("epsilon must be a finite non-negative number")
41
42          # Find the index where epsilon would be inserted to maintain sorted order.
43          # partition_point finds the first position where epsilon <= e.
44          idx = bisect.bisect_right(epsilon_grid, epsilon)  #
45
46          if idx == 0:  #
47              return 1.0
48          else:
49              return delta_grid[idx - 1]
50
51      return PrivacyProfile(profile, epsilon_grid=epsilon_grid)
```

Postcondition

Theorem 1.1. The pseudocode returns a privacy profile $p : \mathbb{R}_{\geq 0} \to [0, 1]$ such that:

1. For $\epsilon < \epsilon_{\min}$ (the smallest sampled epsilon), $p(\epsilon) = 1$.
2. For $\epsilon \geq \epsilon_{\min}$, $p(\epsilon) = \delta_i$ where i is the largest index such that $\epsilon_i \leq \epsilon$.
3. The profile is conservative: returned δ values are at least as large as any valid delta at that epsilon.

Proof. Let $\sigma = \text{sort}(\text{pairs})$ be the list after sorting by ϵ ascending and δ descending (line 15). Let $\sigma' = \text{dedup}(\sigma)$ be the result after removing duplicates on ϵ (lines 18–26), retaining the maximum δ for each unique ϵ. Let $\epsilon_1, \ldots, \epsilon_n$ and $\delta_1, \ldots, \delta_n$ be the epsilon and delta values in σ'.

Sorting and Deduplication Sorting by $(\epsilon, -\delta)$ ensures that among all pairs with the same ϵ, the one with the largest δ appears first. Since duplicates are removed greedily, only the maximum δ per unique ϵ survives deduplication. This is conservative: for any two profiles at the same ϵ with deltas δ and δ' where $\delta > \delta'$, the profile with δ is weaker (more private).

Monotonicity Invariant Let I be the property: $\delta_i \geq \delta_{i+1}$ for all $1 \leq i < n$. The monotonicity check (line 32) ensures I holds. If I fails, an error is returned. Property I is necessary for the result to be a valid privacy profile: as privacy budget (epsilon) increases, privacy guarantees (delta) cannot improve.

Correctness of Query Evaluation For a query $\epsilon \in [0, \infty)$, let $i^* = \max\{i : \epsilon_i \leq \epsilon\}$ (or $i^* = 0$ if no such i exists). The binary search (line 44) returns i^*, and the profile returns (lines 46):

$$p(\epsilon) = \begin{cases} 1 & \text{if } i^* = 0 \\ \delta_{i^*} & \text{otherwise} \end{cases}$$

This is the unique staircase profile on σ'.

Conservatism The profile is conservative by design:

- For $\epsilon < \epsilon_1$, returning $p(\epsilon) = 1$ is the worst-case bound, as all valid privacy profiles are bounded by 1.
- For $\epsilon \geq \epsilon_1$, the staircase property means $p(\epsilon)$ is the δ value associated with the largest sampled point not exceeding ϵ. Any interpolation between sampled points would necessarily weaken privacy guarantees.

$\square$